

linear_algebra

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1 A Crash Course in Linear Algebra

{note} This notebook is a standalone, self-paced primer on the linear algebra used in optimization: vectors and matrices; matrix operations (scalar, dot, and matrix-vector products); determinants and inverses - including cofactor expansion and the general Gauss-Jordan elimination method, which works for matrices of any size; solving linear systems; and gradients, Hessians, and definiteness. All computations are worked entirely by hand - no programming is required.

Learning Objectives

By the end of this notebook, you will be able to: 1. Represent optimization problems using vector and matrix notation, and perform scalar, dot, and matrix-vector products by hand. 2. Compute determinants and inverses of matrices — including matrices larger than 2×2 , using cofactor expansion and the general Gauss-Jordan elimination method — and use the inverse to solve systems of linear equations by hand. 3. Compute gradients and Hessians of multivariable functions, and classify stationary points using the eigenvalues of the Hessian. 4. Verify each hand computation using an independent check — substitution, the definition of an inverse, or the definition of an eigenvalue.

Prerequisites: None — this is a standalone primer requiring only basic algebra.

Estimated time: 60–90 minutes.

1.1 Intuition

Why linear algebra matters: Optimization problems are routinely written in vector and matrix notation — a linear program’s objective and constraints can be packed into a single compact statement, $\max \mathbf{c}^\top \mathbf{x}$ subject to $\mathbf{Ax} \leq \mathbf{b}$, no matter how many decision variables or constraints it involves. Many non-linear solution methods (gradient descent, Newton’s method, interior-point methods) likewise move through $\mathbb{R}^n$ guided by gradients and Hessians. This notebook builds the vocabulary and by-hand computational skills to read and use both.

1.2 Vectors and Matrices

Symbol	Meaning	Example
$\mathbf{x} \in \mathbb{R}^n$	column vector of n numbers	$\mathbf{x} = (5, 8)^\top \in \mathbb{R}^2$
$\mathbf{A} \in \mathbb{R}^{m \times n}$	matrix with m rows, n columns	$\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \in \mathbb{R}^{2 \times 2}$, with $a_{21} = 1$
$\mathbf{0}, \mathbf{I}$	zero vector/matrix, identity matrix	stay bold throughout this notebook, even where other symbols later drop bold (see the notation-switch note below)

1.3 Matrix Operations

1.3.1 Multiplication

- **Scalar Product:** $k\mathbf{A}$ — multiply every entry of $\mathbf{A}$ by the scalar k .

$$k\mathbf{A} = k \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} ka & kb \\ kc & kd \end{bmatrix}$$

$$\text{Solve for: } \mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}; k = 3$$

$$k\mathbf{A} = \begin{bmatrix} 3(2) & 3(1) \\ 3(1) & 3(3) \end{bmatrix} = \begin{bmatrix} 6 & 3 \\ 3 & 9 \end{bmatrix}$$

- **Dot Product:** $\mathbf{c}^\top \mathbf{x}$ — multiply corresponding entries and sum.

$$\mathbf{c}^\top \mathbf{x} = \sum_{i=1}^n c_i x_i = c_1 x_1 + c_2 x_2 + \cdots + c_n x_n$$

$$\text{Solve for: } \mathbf{c} = (40, 30)^\top; \mathbf{x} = (5, 8)^\top$$

$$\mathbf{c}^\top \mathbf{x} =$$

- **Matrix-Vector Product:** $\mathbf{Ax}$ — dot each row of $\mathbf{A}$ with $\mathbf{x}$.

$$\mathbf{Ax} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix}$$

$$\text{Solve for: } \mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}, \mathbf{x} = (5, 8)^\top$$

$$\mathbf{Ax} = \begin{bmatrix} 2(5) + 1(8) \\ 1(5) + 3(8) \end{bmatrix}$$

$$= \begin{bmatrix} 18 \\ 29 \end{bmatrix}$$

1.3.2 Determinant

- **2×2 Matrix:** For a 2×2 matrix $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the determinant is given by,

$$\det(\mathbf{A}) = ad - bc$$

$$\begin{aligned} \text{Solve for:} \quad \mathbf{A} &= \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \\ \det(\mathbf{A}) &= 2(3) - 1(1) \\ &= 5 \end{aligned}$$

- **3×3 Matrix:** For a 3×3 matrix, expand along the first row: each entry is multiplied by the determinant of the 2×2 matrix left after deleting its row and column, with alternating signs.

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

$$\begin{aligned} \text{Solve for:} \quad \mathbf{A} &= \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix} \\ \det(\mathbf{A}) &= 1(1(0) - 4(6)) - 2(0(0) - 4(5)) + 3(0(6) - 1(5)) \\ &= 1(-24) - 2(-20) + 3(-5) \\ &= -24 + 40 - 15 \\ &= 1 \end{aligned}$$

- **$n \times n$ Matrix:** The same idea generalizes to any size via **cofactor (Laplace) expansion** along any row or column: multiply each entry by the determinant of the $(n-1) \times (n-1)$ **minor** left after deleting its row and column, apply alternating signs, and sum. Here, $\mathbf{M}_{1j}$ is the $(n-1) \times (n-1)$ minor obtained by deleting row 1 and column j from $\mathbf{A}$. This recursively reduces an $n \times n$ determinant to a sum of $(n-1) \times (n-1)$ determinants, so it can be expanded along any row or column — but the number of terms grows quickly with n , which is why larger determinants are computed via row reduction (as the product of the pivots) rather than by hand with cofactors.

$$\det(\mathbf{A}) = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(\mathbf{M}_{1j})$$

1.3.3 Inverse

- **2×2 Matrix:** For a 2×2 matrix $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is given as

$$\mathbf{A}^{-1} = \frac{1}{\det(\mathbf{A})} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Solve for: $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$

$$\begin{aligned} \mathbf{A}^{-1} &= 1/5 \times \begin{bmatrix} 3 & -1 \\ -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 0.6 & -0.2 \\ -0.2 & 0.4 \end{bmatrix} \end{aligned}$$

- **$n \times n$ Matrix:** The closed-form formula above only works for 2×2 matrices. For any square invertible matrix, the general method is to augment it with the identity matrix, $[\mathbf{A} \mid \mathbf{I}]$, and apply row operations (swap two rows, scale a row, or add a multiple of one row to another) until the left block becomes $\mathbf{I}$. The right block is then $\mathbf{A}^{-1}$.

Solve for:

$$\mathbf{A} = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix}$$

Step 0: Setup

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 5 & 6 & 0 & 0 & 0 & 1 \end{array} \right]$$

Step 1: Row 3 $\rightarrow$ Row 3 $-5 \times$ Row 1

$$\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & -4 & -15 & -5 & 0 & 1 \end{array} \right]$$

Step 2: Row 1 $\rightarrow$ Row 1 $-2 \times$ Row 2; Row 3 $\rightarrow$ Row 3 $+4 \times$ Row 2

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -5 & 1 & -2 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array} \right]$$

Step 3: Row 1 $\rightarrow$ Row 1 $+5 \times$ Row 3; Row 2 $\rightarrow$ Row 2 $-4 \times$ Row 3

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -24 & 18 & 5 \\ 0 & 1 & 0 & 20 & -15 & -4 \\ 0 & 0 & 1 & -5 & 4 & 1 \end{array} \right]$$

$$\mathbf{A}^{-1} = \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix}$$

Solving Linear Systems Given $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{A}$ invertible, the solution is $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$

Solve for:

$$x_1 + 2x_2 + 3x_3 = 11$$

$$x_2 + 4x_3 = 9$$

$$5x_1 + 6x_2 = 21$$

We have:

$$\begin{aligned}\mathbf{A} &= \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{bmatrix} & \mathbf{b} &= \begin{bmatrix} 11 \\ 9 \\ 21 \end{bmatrix} \\ \mathbf{x} &= \begin{bmatrix} -24 & 18 & 5 \\ 20 & -15 & -4 \\ -5 & 4 & 1 \end{bmatrix} \begin{bmatrix} 11 \\ 9 \\ 21 \end{bmatrix} = \begin{bmatrix} -264 + 162 + 105 \\ 220 - 135 - 84 \\ -55 + 36 + 21 \end{bmatrix} \\ &= \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}\end{aligned}$$

1.4 Gradients, Hessians, and Definiteness

Notation switch: From here on, vectors and matrices drop the boldface for brevity - $\mathbf{x}$ instead of $\mathbf{\mathbf{x}}$. Only the zero vector $\mathbf{0}$ and identity matrix $\mathbf{I}$ stay bold, exactly as above.

1.4.1 Gradient

The gradient $\nabla f(x)$ is the vector of partial derivatives of f — it points in the direction of steepest increase, and $\nabla f(x^*) = \mathbf{0}$ at a stationary point.

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$$

Solve for: $f(x_1, x_2) = (x_1 - 2)^2 + (x_2 - 3)^2$, $x_0 = (0, 0)$

$$\frac{\partial f}{\partial x_1} = 2(x_1 - 2)$$

$$\frac{\partial f}{\partial x_2} = 2(x_2 - 3)$$

$$\begin{aligned}\nabla f(x_0) &= \begin{bmatrix} 2(0 - 2) \\ 2(0 - 3) \end{bmatrix} \\ &= \begin{bmatrix} -4 \\ -6 \end{bmatrix}\end{aligned}$$

1.4.2 Hessian

The Hessian $\nabla^2 f(x)$ is the matrix of second partial derivatives of f .

$$\nabla^2 f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

$$\text{Solve for: } f(x_1, x_2) = (x_1 - 2)^2 + (x_2 - 3)^2$$

$$\frac{\partial^2 f}{\partial x_1^2} = 2$$

$$\frac{\partial^2 f}{\partial x_2^2} = 2$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = 0$$

$$\nabla^2 f(x) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

1.4.3 Definiteness

For any size, solve the characteristic equation $\det(H - \lambda \mathbf{I}) = 0$ for λ ; the signs of the eigenvalues classify definiteness directly (all positive: positive definite; all negative: negative definite; mixed signs: indefinite).

$$\det(H - \lambda \mathbf{I}) = 0$$

$$\text{Solve for: } H = \begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\text{We have: } \det(H - \lambda \mathbf{I}) = 0$$

$$\det\left(\begin{bmatrix} 2 & 1 \\ 1 & -1 \end{bmatrix} - \lambda \mathbf{I}\right) = 0$$

$$\det\left(\begin{bmatrix} 2 - \lambda & 1 - \lambda \\ 1 - \lambda & -1 - \lambda \end{bmatrix}\right) = 0$$

$$(2 - \lambda)(-1 - \lambda) - 1(1) = 0$$

$$\lambda^2 - \lambda - 3 = 0$$

$$\lambda = \frac{1 \pm \sqrt{13}}{2}$$

$$\lambda_1 \approx 2.30, \quad \lambda_2 \approx -1.30$$

One positive and one negative eigenvalue confirm H is indefinite.

1.5 Take-Away Exercises

Complete the following exercises on your own, showing your work by hand.

Exercise 1 — Determinant of a 2×2 Matrix Let $\mathbf{B} = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$.

Your Tasks

- A. Compute $\det(\mathbf{B})$ using the 2×2 formula.
- B. State whether $\mathbf{B}$ is invertible, and explain why.

Exercise 2 — Determinant of a 3×3 Matrix Let $\mathbf{M} = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 3 & 2 \\ 0 & 1 & 1 \end{bmatrix}$.

Your Tasks

- A. Compute $\det(\mathbf{M})$ by cofactor expansion along the first row.
- B. State whether $\mathbf{M}$ is invertible, and explain why.

Exercise 3 — Inverse of a 2×2 Matrix Let $\mathbf{C} = \begin{bmatrix} 3 & 1 \\ 2 & 1 \end{bmatrix}$.

Your Tasks

- A. Compute $\mathbf{C}^{-1}$ using the direct 2×2 formula.
- B. Verify your answer by confirming $\mathbf{C}\mathbf{C}^{-1} = \mathbf{I}$.

Exercise 4 — Inverse of a 3×3 Matrix Let $\mathbf{D} = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$.

Your Tasks

- A. Compute $\mathbf{D}^{-1}$ using Gauss-Jordan elimination.
- B. Verify your answer by confirming $\mathbf{D}\mathbf{D}^{-1} = \mathbf{I}$.

Exercise 5 — Solve a 3×3 System A distributor blends three grades of aggregate to meet three targets: $2x_1 + x_2 + x_3 = 8$ (total volume, in tonnes), $x_1 + 3x_2 + 2x_3 = 11$ (a strength index), and $3x_1 + x_2 + 4x_3 = 19$ (a cost index).

Your Tasks

- A. Write the system in matrix form $\mathbf{Ax} = \mathbf{b}$.
- B. Solve for x_1, x_2, x_3 by hand, using either row-reduction or the matrix inverse.
- C. Verify your solution by substituting back into all three original equations.

Exercise 6 — Gradient of a Three-Variable Function Let $f(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 3x_3^2 + x_1x_2 + x_2x_3$.

Your Tasks

A. Derive the gradient $\nabla f(x)$ by computing all three partial derivatives.

B. Evaluate $\nabla f(x)$ at $x_0 = (1, 1, 1)^\top$.

Exercise 7 — Hessian of a Three-Variable Function Using the same $f(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 3x_3^2 + x_1x_2 + x_2x_3$ from Exercise 6.

Your Tasks

A. Derive the Hessian $\nabla^2 f(x)$ by computing all second partial derivatives.

B. State whether $\nabla^2 f(x)$ depends on x , and explain why.

Exercise 8 — Definiteness of a 3×3 Hessian At a stationary point, the Hessian is $H =$

$$\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Your Tasks

A. Solve the characteristic equation $\det(H - \lambda \mathbf{I}) = 0$ for the eigenvalues (expand along the first row to simplify).

B. Classify the stationary point based on the signs of the eigenvalues.

1.6 Further Reading

- *Strang, G.*, “Introduction to Linear Algebra,” Wellesley-Cambridge Press — the standard reference for the vector/matrix material in this notebook.
- *Strang, G.*, MIT OpenCourseWare 18.06 “Linear Algebra,” <https://ocw.mit.edu/courses/18-06-linear-algebra-spring-2010/> — free video lectures covering the same material in depth.
- *Boyd, S. and Vandenberghe, L.*, “Introduction to Applied Linear Algebra,” <https://web.stanford.edu/~boyd/vmls/> — a free, application-driven text covering vectors, matrices, and least squares.
- *Nocedal, J. and Wright, S.*, “Numerical Optimization,” Springer — the standard reference for gradients, Hessians, and definiteness in the context of optimization algorithms.